

(a)

Workflow of evo3D

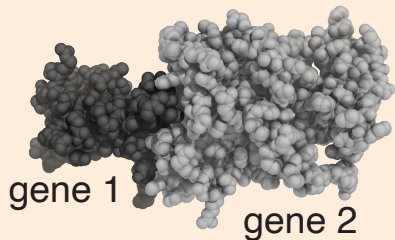
User inputs

1. Multiple Sequence Alignment(s) (msa_file)

```
TATCAAGAAGTTCATGGTGT
TATCCAGAAGATCATGGTGT
TATCAAGA---ACATGGCGC
TACTCAAGA---TCATGGTGC
TATTCGAGAAGTTCATGGTGT
```

... gene 1
... gene 2

2. PDB/mmCIF(s) (pdf_file)



```
run_evo3d(msa_file, pdb_file, chain = 'auto')
```

```
msa_to_ref() + pdb_to_patch()
```

```
aln_msa_to_pdb()
```

```
collapse_to_codon()
```

```
calculate_patch_stats()
```

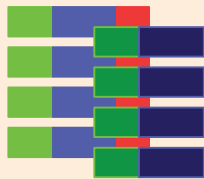
Internal evo3D modules

run_evo3d() outputs

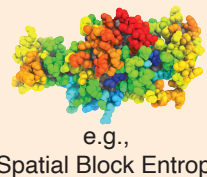
MSA-PDB Table

chain	msa_index	ref_aa	ref_nt	patch_index	patch_aa	patch_nt
chain_1	1	A	ATG	1	A	ATG
chain_1	2	T	ACT	2	T	ACT
chain_1	3	C	CGT	3	C	CGT
chain_1	4	A	AGT	4	A	AGT
chain_1	5	G	GAT	5	G	GAT
chain_1	6	T	TGT	6	T	TGT
chain_1	7	T	TGT	7	T	TGT
chain_1	8	C	CGT	8	C	CGT
chain_1	9	A	AGT	9	A	AGT
chain_1	10	G	GAT	10	G	GAT
chain_1	11	T	TGT	11	T	TGT
chain_1	12	T	TGT	12	T	TGT
chain_1	13	C	CGT	13	C	CGT
chain_1	14	A	AGT	14	A	AGT
chain_1	15	G	GAT	15	G	GAT
chain_1	16	T	TGT	16	T	TGT
chain_1	17	T	TGT	17	T	TGT
chain_1	18	C	CGT	18	C	CGT
chain_1	19	A	AGT	19	A	AGT
chain_1	20	G	GAT	20	G	GAT

Spatial Haplotypes

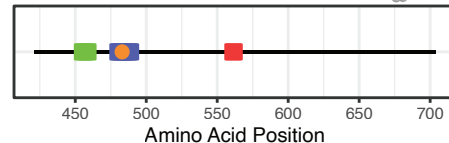
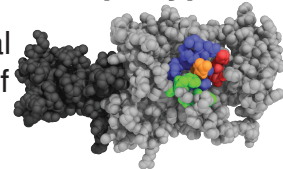


Selection Maps



(b) Extracting Spatial Haplotypes

1. Locate the spatial neighbourhood of residue 483



2. Propagate amino acid positions to codon positions



3. Extract and combine MSA subsets

